Recording

EN 580.133
mRNA as Therapeutics (Part III)

Elaine Zhelan Chen

Jan 8, 2025

Created by Elaine Zhelan Chen

Final Presentation Teams

- Presentation order
  - Team 3 -> Team 2 -> Team 4

```
> set.seed(154)
> random_integers <- sample(2:4, 3, replace = FALSE)
> print(random_integers)
[1] 3 2 4
>
```

	Team #1	Team #2	Team #3	Team #4
	mRNA-Propionic Acidemia (PA)	mRNA-Melanoma	siRNA-Hyperlipidemia	siRNA-Hemophilia
Team member 1		Edward Li, eli36@jh.edu	Emma Fredman, efredma1@jh.edu	Resham Talwar, rtalwar2@jh.edu
Team member 2		Habikah Baldeh, hbaldeh1@jh,edu	Alex Flynn, aflynn18@jh.edu	Aayushi Sachdeva, asachde9@jh.edu
Team member 3		Jonathan Xiao, Jxiao30@jh.edu	Sunny Li, jli409@jh.edu	Flora Tao, ztao13@jh.edu
Team member 4		Melissa Martinez mmart213@jhu.edu	Kelsey Liggett, kligget2@jh.edu	Nicolas Poliak, npoliak3@jh.edu

Recap

- Strategies for achieving targeted delivery in a specific tissue or cell type
  - LNP surface conjugates
  - Net charge of LNPs by tuning the lipid composition
  - Route of administration
- ssRNA and dsRNA can activate the innate immune response
  - ssRNA, incorporation of modified nucleotides
  - dsRNA, purification by cellulose chromatography or HPLC
- Enhance mRNA translation efficiency and stability by sequence/codon optimization
- Manufacture mRNA using in vitro transcription, capping and tailing

Immunogenic and non-immunogenic mRNA therapeutics

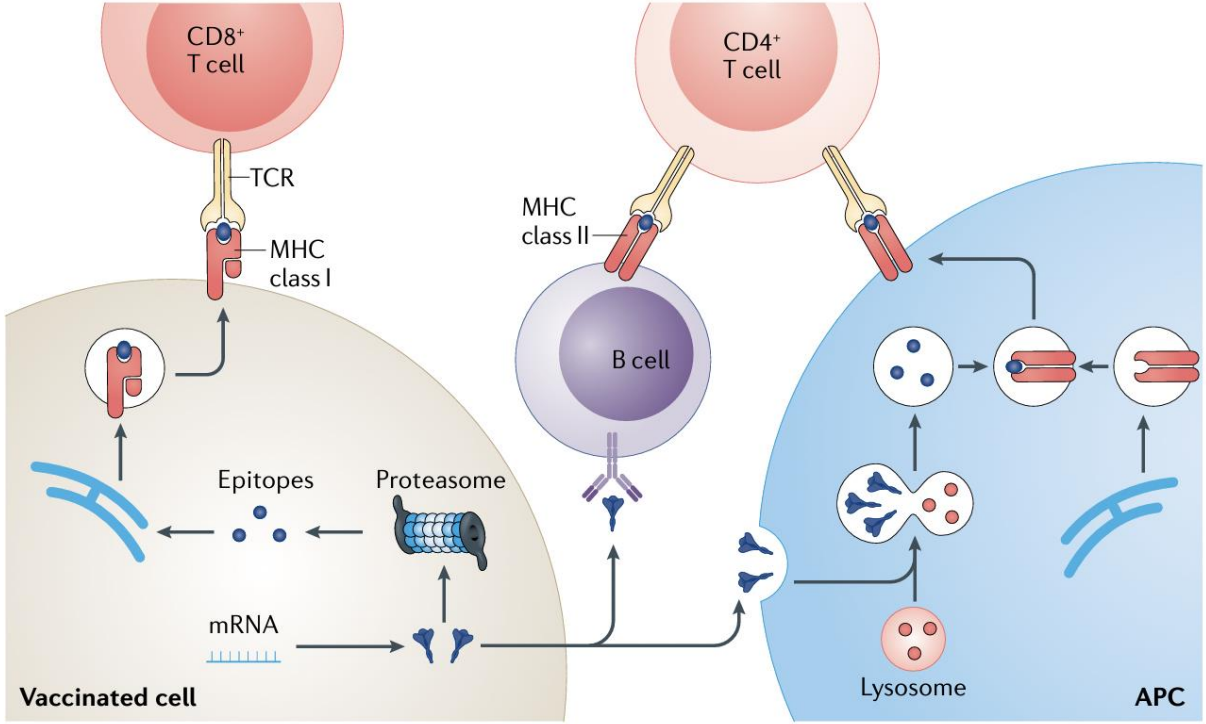
- Immunogenic
  - Prophylactic vaccines
  - Individualized neoantigen therapies (INTs)
- Non-immunogenic
  - Protein replacement therapies

mRNA vaccine

Vaccines

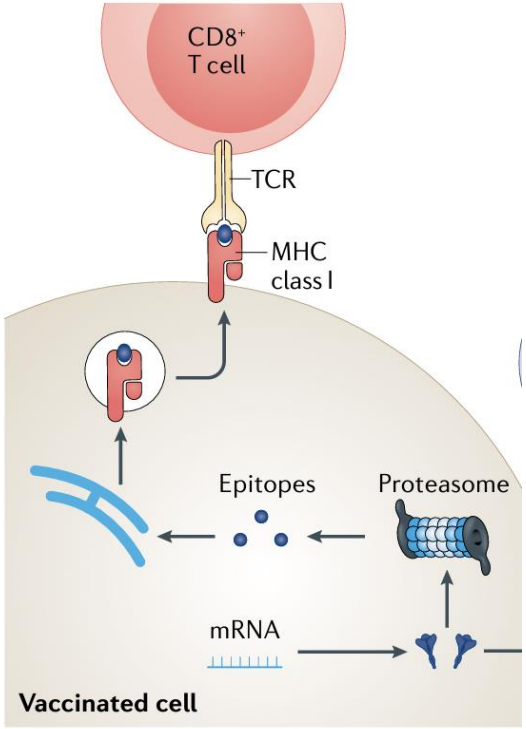
- Vaccines serve the purpose to prevent disease
  - prophylactic vaccine
- Traditional methods of vaccination - Attenuated pathogen
  - Inactivated version or active but weakened version of a pathogen
- Mechanism of vaccines in general
  - Expose our immune systems to a new infectious agent to develop an adaptive immune response

What is the mechanism of mRNA vaccines?



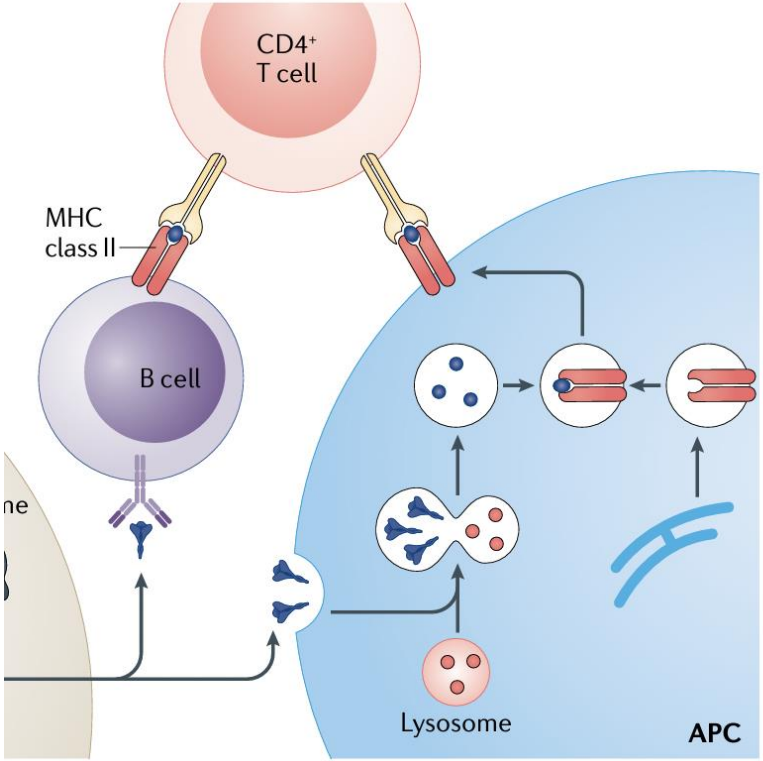
What is the mechanism of mRNA vaccines?

- LNP–mRNA are internalized by somatic cells
- For intramuscular delivery, LNP-mRNA are internalized by muscle cells and antigen-presenting cells (APCs)
- Spike antigens expressed in the cytoplasm are degraded by proteasomes
- Major histocompatibility complex (MHC) class I presents the epitopes to CD8+ T cells
- CD8+ T cells recognize non-self epitopes, become activated, and initiates a cytotoxic response to kill the infected cells



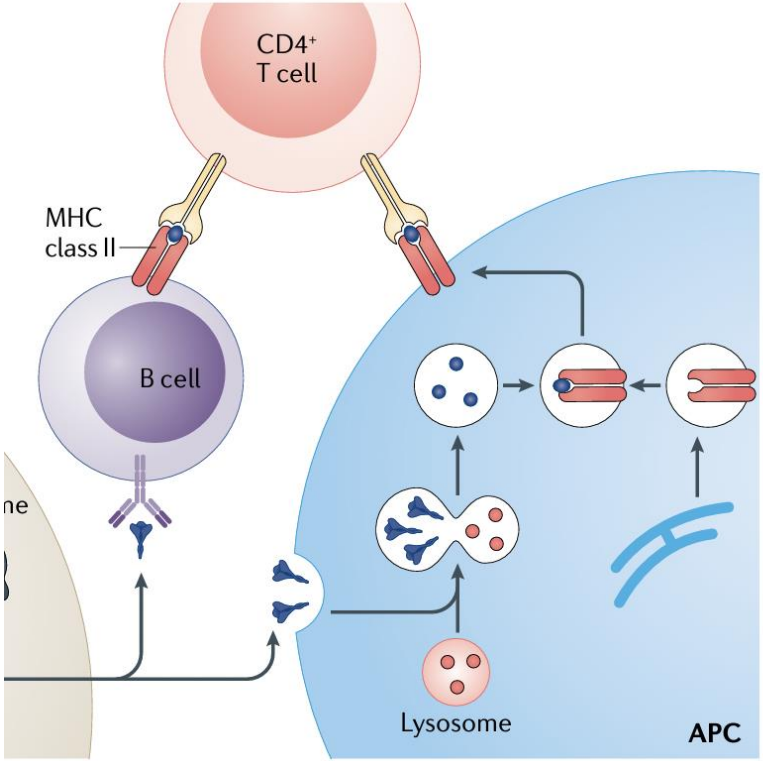
Antigen presenting cells (APCs)

- APCs can engulf pathogens, degrade them in the lysosomes, and present these epitopes using MHC class II molecules
- CD4+ T cells recognize non-self epitopes, become activated, and release cytokines
- Cytokines serve as a signal for other immune cells to start immune response
- CD4+ T cells can also become memory cells to initiate a faster and more robust response when encountering the same pathogen



B Cell response

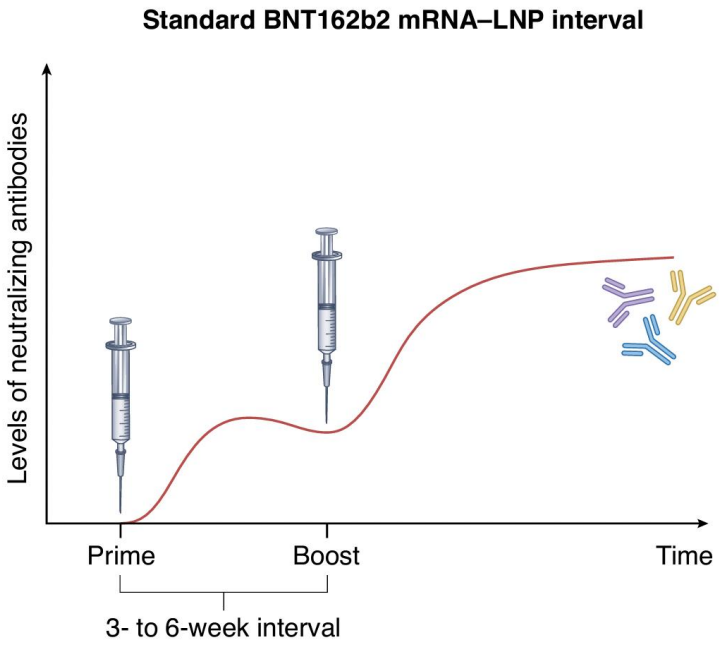
- Secreted antigens can also be internalized by B cell receptors and processed for presentation to CD4+ T cells by MHC class II molecules
- Subsequently, B cells are activated and will proliferate and differentiate into plasma cells
- Plasma cells will start to mass produce antibodies specific to the antigen
- Antibodies will be released into the blood stream and will be readily available to neutralize the pathogen therefore creating immune protection



Immunization duration

- A prime dose and a booster dose
- Boosters can increase neutralizing antibody levels
- Important for protecting elderly and immunocompromised individuals

- Two doses of mRNA-1273 also elicited durable antibody responses over a period of 6 months
- The number of antibodies declined slightly over the duration
- However, the neutralizing ability was retained



Emerging Viral Variants

- During viral replication within a host, small errors or mutations can occur in the viral genome
- A viral variant is a strain of the virus that has one or more mutations compared to the original strain of virus
- Mutations can be beneficial to the virus
  - Better immune evasion
  - More efficient transmission
- COVID-19 vaccines reported to be less effective for the later Omicron variant

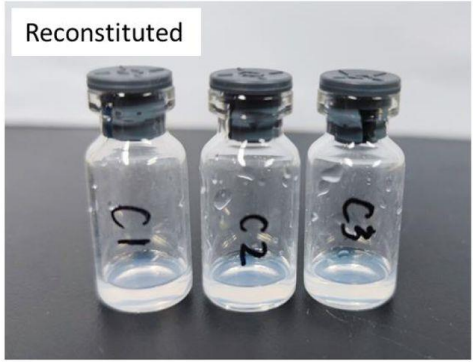
Advantages of mRNA vaccines

- Fast production cycle
  - Match the circulating variant at the time of vaccination
  - High efficacy
- Small footprint
  - 5-liter bioreactor can produce almost a million mRNA vaccine doses in a single reaction
  - Local manufacturing and distribution
- Same ingredients regardless of the sequence
- Quick response to a new emerging pandemic

Disadvantages of mRNA vaccines

- Thermostability
  - Pfizer–BioNTech BNT162b2 requires storage at -60 C in ultra-cold condition
  - Vaccine accessibility can be limited by the cold storage requirements

  - Lyophilized vaccines can be stored at room temperature for over months
    - Reconstitution on site is needed
  - Moderna mRNA-1273 can be stored at 4-8 C for a month and 12 h at room temperature



COVID Vaccine – developing at warp speed

- For reference,
  - MERS
    - MERS-CoV (outbreak in 2012) vaccines required about 2 years of development before phase I trials
    - No MERS-CoV vaccine has been approved yet
  - Mumps
    - The mumps vaccine required 4 years from conception to approval
    - Was then the fastest developed vaccine

COVID Vaccine – developing at warp speed

- The Moderna and Pfizer–BioNTech vaccines
  - Approved 11 months after initiation
  - 8 months oh which were spent in clinical trials

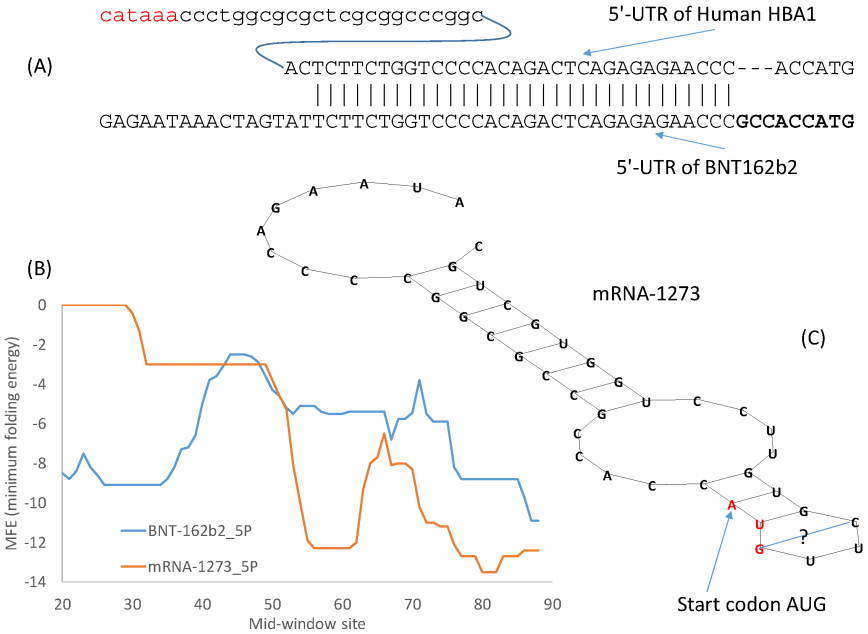
- Pfizer–BioNTech vaccine BNT162b2 received emergency authorization from the FDA on December 11th, 2020
  - the first mRNA drug approved for use in humans
- Moderna vaccine mRNA-1273 was authorized one week later
- Both showed over 90% efficacy in protection against COVID-19

mRNA design

- Components of the mRNA molecule can be individually optimized to enhance RNA stability and translation efficiency
  - 5’ cap
  - 5’ UTR
  - ORF
  - 3’ UTR
  - Poly(A) tail

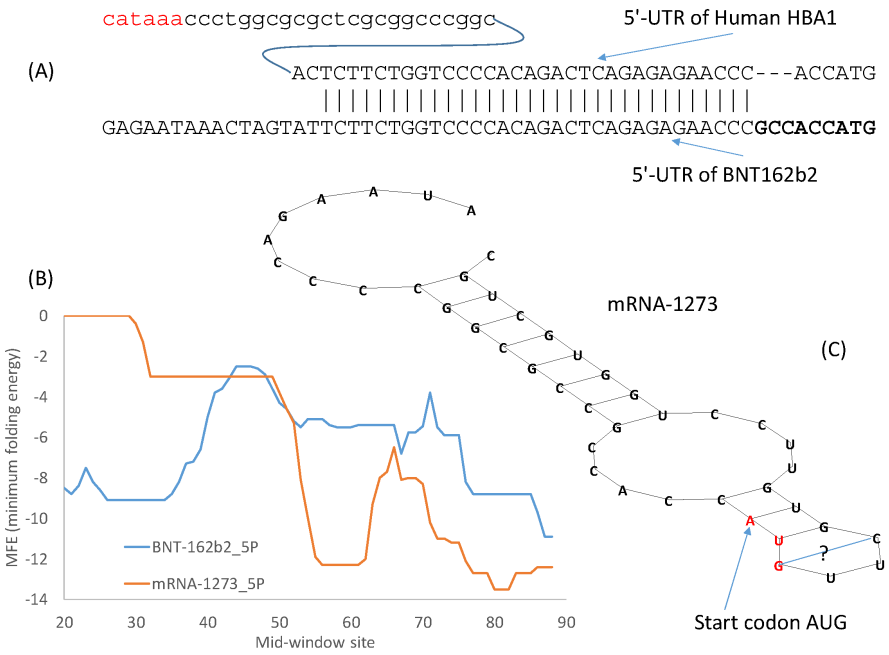
5’ UTR

- 5′ UTR serves two key functions
  - to stabilize mRNA
  - to facilitate scanning by small ribosome subunit to localize the start codon
- Pfizer/BioNTech approach
  - incorporating the 5′ UTR of human α-globin gene
  - HBA1 is a highly expressed human gene, indicating that it is translated very efficiently
- Moderna’s approach
  - a GC-rich element CCCCGGCGCC just before the start codon
  - the secondary structure may increase mRNA stability and translation accuracy by reducing leaky scanning by the ribosome



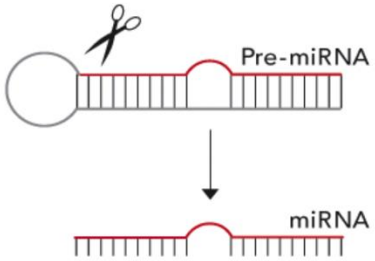
5’ UTR

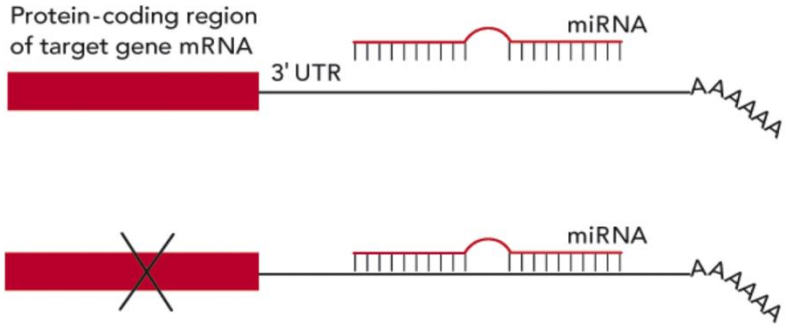
- The Pfizer/BioNTech 5’ UTR strategy seems to be superior
- Moderna vaccine supplies a high dose of 100 µg/dose
- Pfizer/BioNTech use a lower dose of 30 µg/dose



3’ UTR

- 3′ UTR can also affect mRNA stability and translation
- 3’ UTR has segments with complementarity to microRNAs
- Binding of microRNAs to the 3’ UTR can subject the mRNA to degradation or translation inhibition

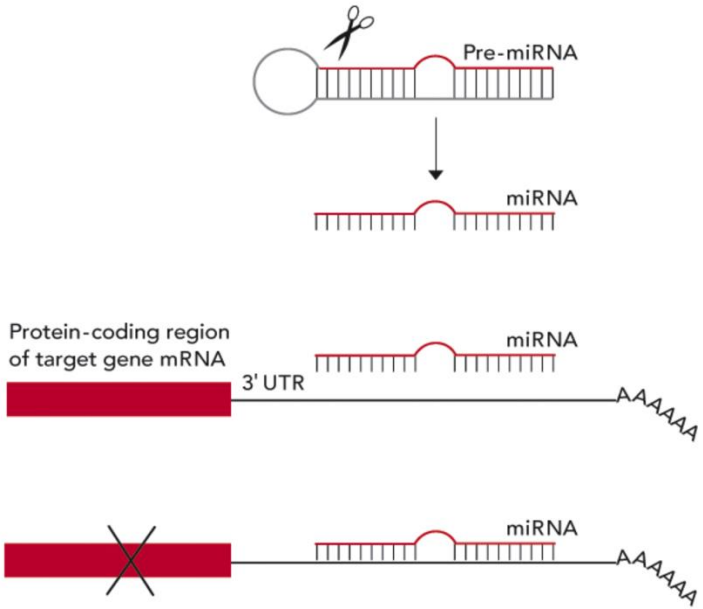




3’ UTR

- Moderna’s approach
  - incorporating the 3’ UTR of human α-globin gene
  - Copy and pasted the 110-nt 3′ UTR of HBA1 between the last stop codon and a poly(A) tail

- Pfizer/BioNTech’s approach
  - Seek to find better performing 3’ UTR than the human β-globin mRNA 3’ UTR
  - Lowest predicted microRNA binding sites
  - human AES/TLE5 gene 3’ UTR followed immediately by human 12S rRNA (mtRNR1) gene 3’ UTR



Poly(A) tail

- Adding poly(A) tail post IVT tends to lead to a variable tail lengths
- Incorporating the poly(A) tail sequence in the DNA plasmid to minimize tail length variability
- Poly(A) tails of >100 bp are optimal for therapeutic mRNAs
- However, the DNA sequences that encode these long poly(A)s destabilize the DNA plasmids used for transcription
  - A solution is to include a short UGC linker in the poly(A) tail
  - The Pfizer–BioNTech vaccine contains a 10 bp UGC linker to produce the sequence A30(10 bp UGC linker)A70
- one-step mRNA synthesis process

COVID Vaccine Anaphylaxis

- Anaphylaxis is a severe, life-threatening allergic reaction also known as allergic or anaphylactic shock
  - Nausea, vomiting
  - Difficulty breathing
  - Can be life-threatening

- 2-4 folds more than typically seen in traditional vaccines
  - Pfizer–BioNTech BNT162b2: 4.7 per million vaccinations
  - Moderna mRNA-1273: 2.5 per per million vaccinations

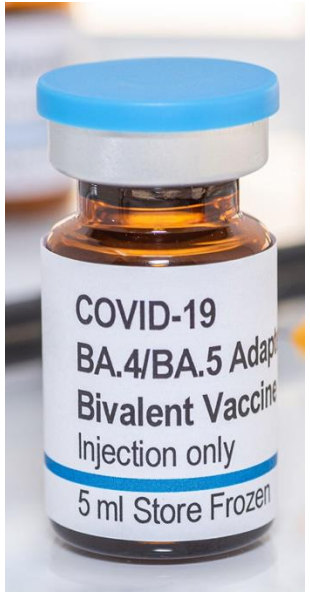
- Potentially due to pre-existing antibodies against the PEG-lipid in the LNPs
  - PEG exist in products like toothpastes, shampoos and laxatives
  - CDC recommends that mRNA vaccines should not be given to individuals with a history of allergic response to any of the components

mRNA vaccine can be multiplexed

- Against multiple infectious agent

- Bivalent COVID vaccine
  - Encoding for both the ancestral WA1/2020 and the Omicron BA.4/BA.5 spike proteins

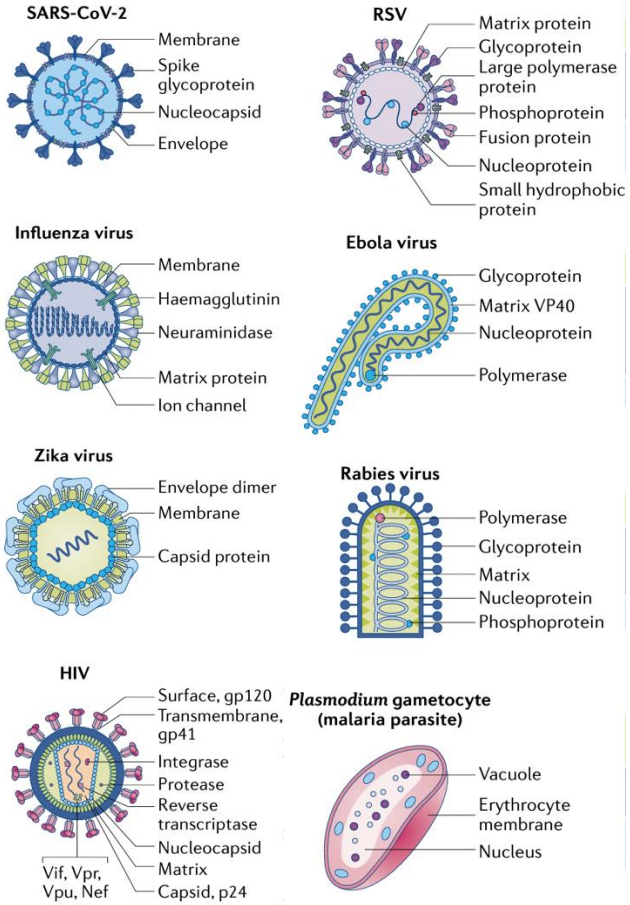
- Quadrivalent seasonal influenza vaccine (mRNA-1010) completed phase 1/2 clinical trial
  - Encoding the hemagglutinin (HA) surface glycoproteins of 4 different strains: A/H1N1, A/H3N2, B/Victoria, and B/Yamagata



mRNA Vaccines outlook

- mRNA vaccines to protect against various pathogens
  - Respiratory Syncytial Virus (RSV) mRNA vaccine approved in 2024
  - Clinical trials of mRNA vaccines for Rabies, HIV-1, Zika, Ebola virus, etc.

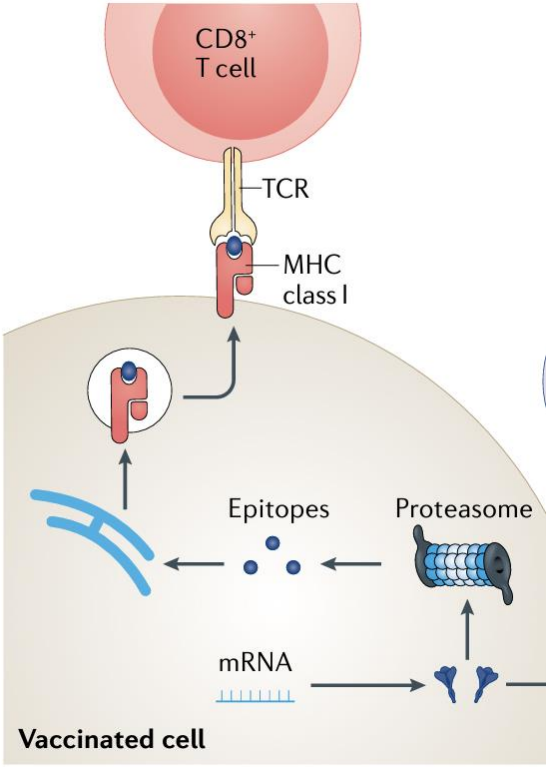
- Pan-coronavirus vaccine - effective at preventing severe disease and/or infection caused by all viruses of the coronavirus family



mRNA neoantigen therapy

mRNA therapeutics to treat cancer

- Tumor cells express mutant proteins that are not present in the host cell which is known as **neoantigens**
- After degradation by the proteasome, these neoantigens are display on the cell surface on MHC class I molecules
- The goal is to train the immune system to recognize an individual's unique cancer and eliminate cancerous cells with cytotoxic T cells



Neoantigens

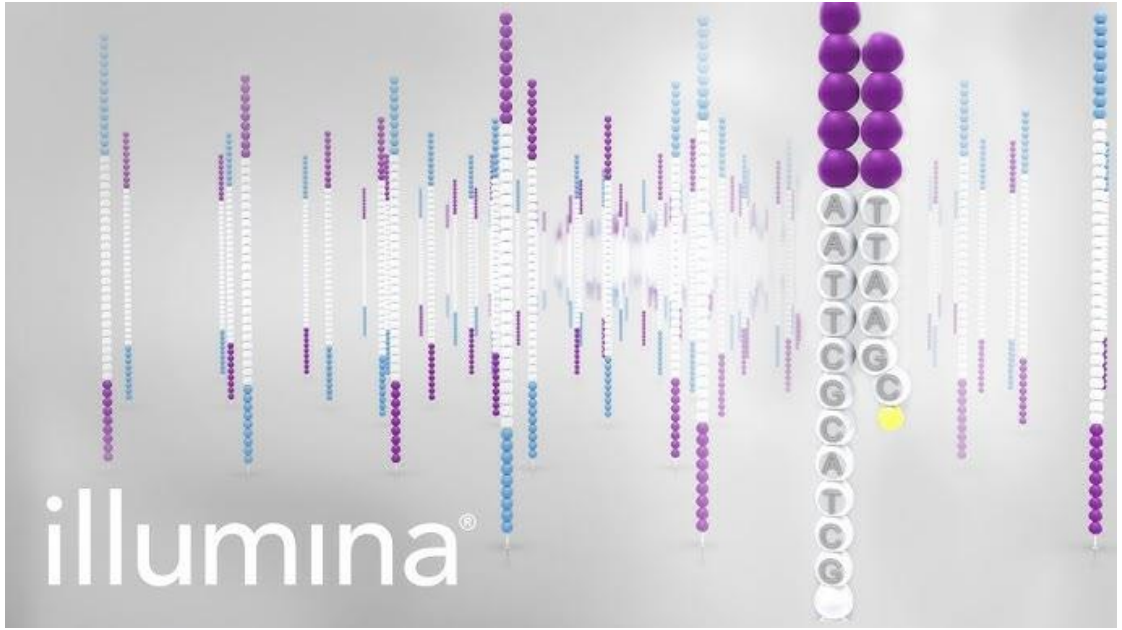
- Can be patient specific
  - Personalized cancer therapy
  - Individualized neoantigen therapies (INTs)

- Aided by
  - Bioinformatic analysis to identify which cancer cell mutations will produce a neoantigen and to predict which neoantigens are most likely to be presented by MHC class I
  - Next generation sequencing to identify unique genetic mutation by comparing the tumor biopsy samples and healthy blood cells

Impact of Next Generation Sequencing (NGS)

- To develop INTs, first all mutations in the cancer cells are identified through deep genome sequencing
- Before the emergence of NGS, sequencing the entire human genome may take over a decade
  - The Human Genome Project took 13 years to complete the sequencing of the human genome
- With NGS, it takes less than 1 day

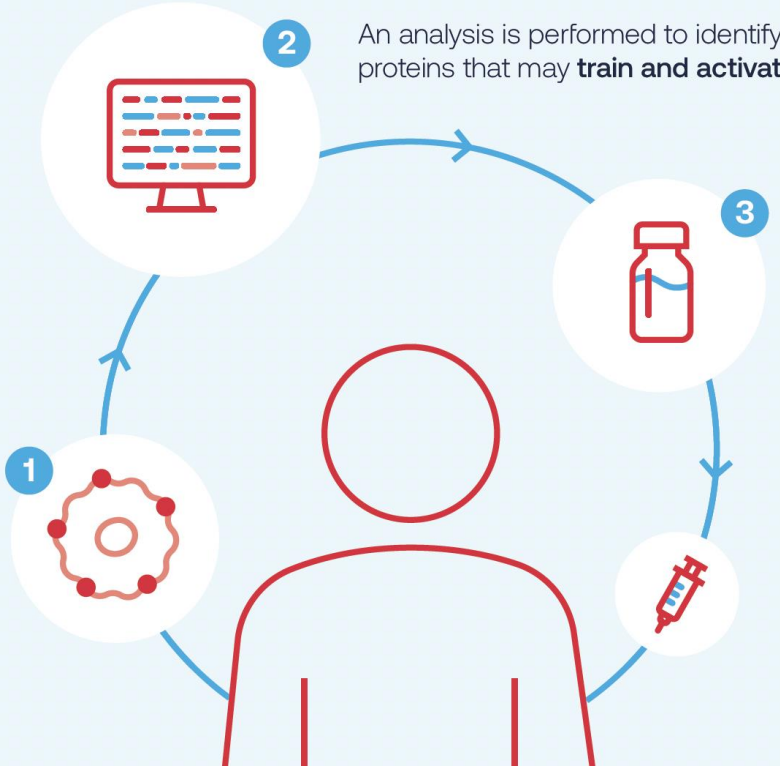
Impact of
Next Generation Sequencing (NGS)



Created by Elaine Zhelan Chen

Personalized mRNA cancer vaccine

How are individualized neoantigen therapies made?



1. First, doctors take a sample of a patient's tumor and blood so they can identify the **unique genetic mutations** of that patient's cancer.

2. An analysis is performed to identify the best cancer fingerprint proteins that may **train and activate** the patient's immune cells.

3. An individualized neoantigen therapy is created for that patient.

   The mRNA in the individualized neoantigen therapy helps show parts of the cancer **fingerprint proteins** to the immune system.

   The patient's immune system may now better **recognize and fight** the patient's cancer.

Created by Elaine Zhelan Chen

Protein replacement therapy

Protein replacement therapy

- Replenish specific protein deficiencies that result either from the protein being absent or non-functional due to inborn mutations
- Can be treated by directly replenishing the proteins
- However, the production of therapeutic proteins is costly and time-consuming

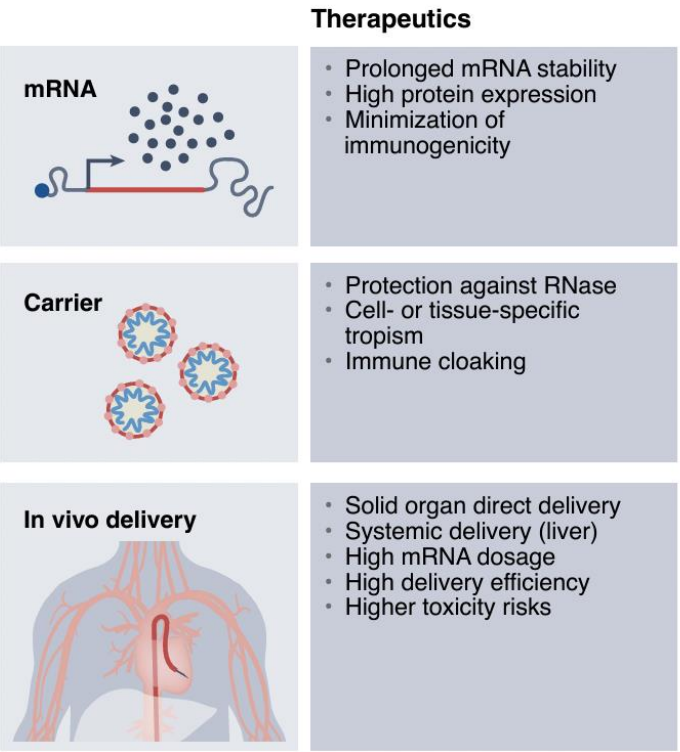
- Key differences from immunogenic therapies
  - **Need 50-1000-fold more mRNA to achieve therapeutic effects!**

Key differences between immunogenic and non-immunogenic mRNA Therapeutics

	Vaccines	Therapeutics
mRNA	• Low protein expression • Immunogenicity favorable	• Prolonged mRNA stability • High protein expression • Minimization of immunogenicity
Carrier	• Protection against RNase • No tropism necessary • Adjuvant immunogenic capacity	• Protection against RNase • Cell- or tissue-specific tropism • Immune cloaking
In vivo delivery	• Local intramuscular delivery • Low mRNA dosage • Low toxicity risks	• Solid organ direct delivery • Systemic delivery (liver) • High mRNA dosage • High delivery efficiency • Higher toxicity risks

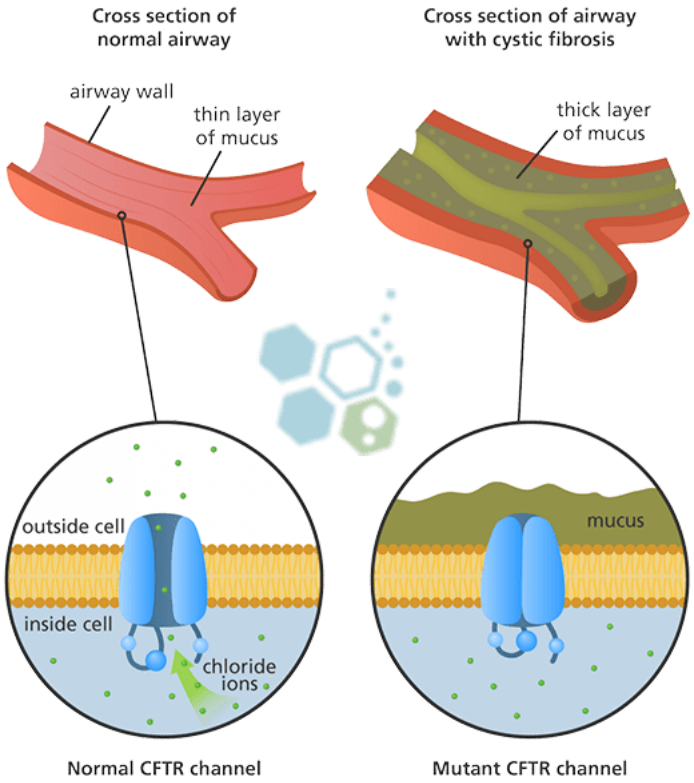
New challenges

- Minimize immunogenicity, specifically minimize delivery to immune cells
- Optimize for cell type/tissue tropism to enhance LNP-mRNA in the target cell type/tissue
- Chronic dosing may be necessary
  - Maintaining robust protein production with repeated dosage over multiple years is challenging
  - Anti-drug antibodies (ADA) may develop which will diminish the therapeutic effect
  - ADAs bind to the drug molecule, neutralize it, and prevent it from interacting with its target
  - ADAs has been seen in some preclinical studies



Cystic fibrosis (CF)

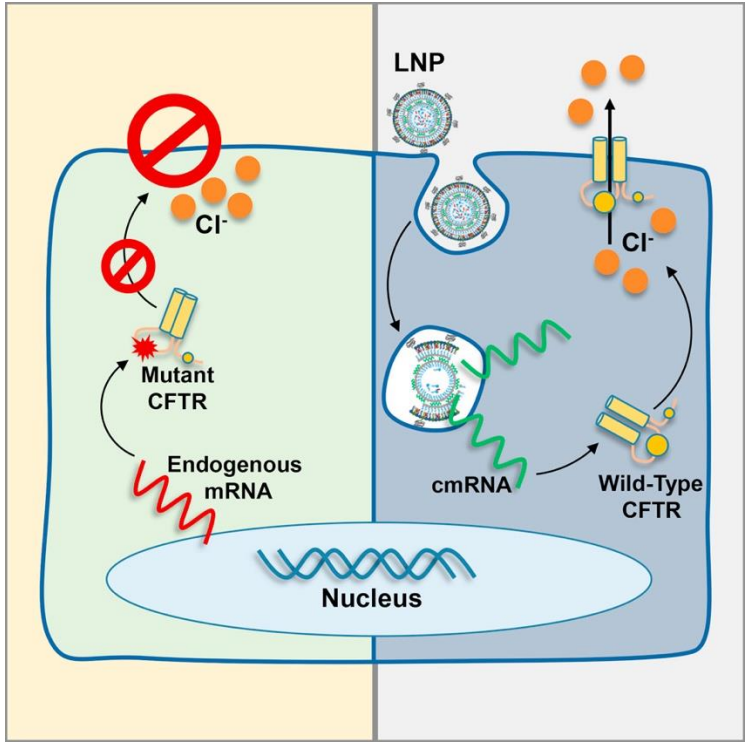
- Cystic fibrosis (CF) is a genetic disease caused by mutations in the cystic fibrosis transmembrane conductance regulator (CFTR) gene
- Mutated CFTR channel cannot pump out chloride ions properly, leading to dehydration outside of the cell
- The mucus in the airways becomes dehydrated and thickens which flatten the cilia
- The cilia can’t move the mucus properly which leads to mucus build-up
- Difficulty in breathing, lung infections, digestion problems



Cystic fibrosis (CF) mRNA therapy

- MRT5005 is a codon-optimized CFTR mRNA delivered by aerosol in lipid nanoparticles was designed as a genotype-agnostic treatment for CF lung disease by Translate Bio

- Phase ½ clinical trial
- From 8 to 24 mg in a single dose, which is a much higher amount compared to 30-100ug per dose in the COVID vaccine
- Generally safe and well tolerated
- Observed fever and hypersensitivity reactions
- Unfortunately, no beneficial effects were observed after the treatment



Problem set #1

- **Due Today, Jan 8th, 5PM EST**

Attendance